

Designing a Seamless VR Experience for Dance Movement Therapy

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Figure 1: We propose *Move & Journey*, a platform designed to reduce stress through improvisational movements in VR. Users can experience an interactive and immersive environment through (left) visual, (middle) auditory, and (right) haptic feedback that responds to their body movements.

ABSTRACT

In our daily lives, we are exposed to various stressors, which, even in mild forms, can lead to serious health issues when prolonged. Dance Movement Therapy (DMT) uses movement as a form of psychotherapy to improve health and well-being, but traditional DMT often requires direct interaction with therapists and physical props, limiting its accessibility for daily use. To address this challenge, we propose *Move & Journey*, a VR application that enables DMT experiences by providing a virtual environment that dynamically adapts to users' movements through visual, auditory, and tactile feedback. This system allows users to engage in DMT alone while maintaining therapeutic principles in an immersive virtual space. Our contributions include empirical insights from the development of dynamic feedback mechanisms for virtual therapy environments and the implementation of features that connect the therapeutic virtual environment and the real world seamlessly.

Index Terms: Virtual reality, Dance movement therapy, Stress reduction

1 INTRODUCTION AND RELATED WORK

In modern society, people face various stressors, such as task deadlines, workplace pressures, and social discomfort, which can trigger negative mental and physical reactions, including frustration, anger, and irritability [5].

Dance Movement Therapy (DMT) is a type of psychotherapy that promotes emotional expression and stress relief, improving mental well-being¹. Emerging studies have shed light on the positive impacts of creative and expressive therapies in the workplace, highlighting reductions in stress levels and improvements in employee

satisfaction and productivity [16, 18]. Dosumu-Lawal et al. conducted a survey of 528 academic staff and concluded that the system works not only as a method for stress relief and emotional regulation but also as a catalyst for enhancing job satisfaction and overall quality of life [3]. While they recommend the implementation of regular DMT sessions facilitated by certified therapists, it is difficult to establish such sessions regularly.

On the other hand, Virtual Reality (VR) has gained attention as a therapeutic tool, demonstrating effectiveness in anxiety management [11], acrophobia treatment [13], and post-traumatic stress disorder (PTSD) therapy [12, 17]. VR offers advantages such as safety, immersion, reproducibility, and control over what users see in the virtual environment (VE), making it a promising alternative or complement to traditional approaches. For example, Tarrant et al. [20] and Kaplan-Rakowski et al. [8] reported that VR-based meditation provides experiences that are equivalent to or even better than traditional meditation methods. They attributed this to the immersive nature of VR environments, which allow users to focus, minimize distractions, and alleviate stress. With the advancements in accessible VR devices and technology, VR is increasingly considered a viable tool for daily stress management.

Previous studies have also explored the potential of Augmented Reality (AR), Mixed Reality (MR), and VR environments for DMT [15]. VR and MR systems have been proposed as personalized therapeutic tools that respond to experts' psychological and physical states. However, these studies have been limited to approaches such as remote DMT using existing VR applications like YouTube VR [1] or DMT designed for children using CAVE-based MR systems [9]. To our knowledge, there has been no research investigating the stress-relief effects of VR-based DMT. To make VR-based DMT more widely accessible, it is necessary to design and evaluate VR environments grounded in the principles of DMT.

The goal of this study is to design a VR application that enables DMT experiences based on real-world therapy principles. Specifically, we conducted interviews and workshops with 10 DMT experts to identify design requirements for implementing DMT in

¹<https://adta.memberclicks.net/what-is-dancemovement-therapy>

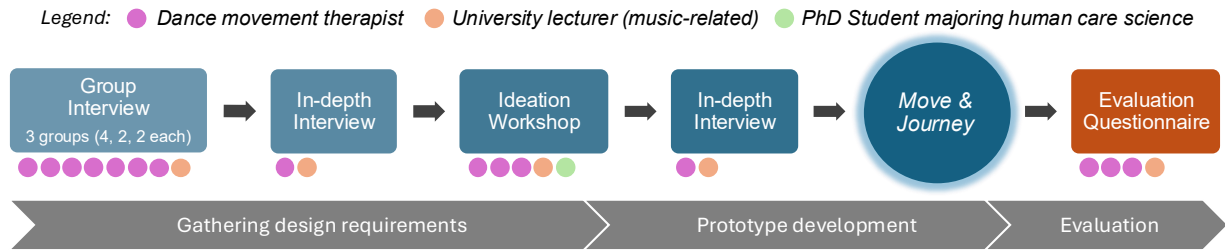


Figure 2: Developing *Move & Journey* through an iterative process.

VE. Based on the insights gained, we focused on three key design elements: *Seamless Session Flow*, *Immersive Imaginative Space*, and *Movement-Based Multisensory Feedback*. Using these design requirements, we propose *Move & Journey*, a platform designed to reduce stress through improvisational movements in VR which aims to recreate the experience of real-world DMT and offers a novel approach to daily stress management.

This work contributes to empirical insights into the design considerations and potential of VR-based DMT, along with the implementation of seamless session flows and multisensory feedback integration.

2 ITERATIVE DESIGN PROCESS

We derived the design requirements for implementing *Move & Journey*, a VR system prototype aimed at reducing stress through improvisational movements. This was achieved through an iterative process [14] involving a series of activities with DMT experts—see Figure 2 for details on the activities and participants. This process aimed to (1) gather insights into the characteristics and limitations of current DMT practices, (2) introduce immersive technologies to experts, and (3) prototype ideas for integrating DMT into VR. This process involved 10 DMT experts affiliated with the Japan Dance Therapy Association². They work in various settings, including psychiatric hospitals, day care centers, and elderly care facilities. Due to geographical constraints, all activities were conducted online. The interviewer remotely demonstrated the VR system while participants observed the process and engaged in discussions.

First, during the group interview, we interviewed eight experts to identify challenges and practices in current therapy sessions. We then implemented a technology probe [7] using VR materials inspired by previous studies to explore different features offered by immersive technologies, such as avatars, particles, motion trails, or terrain changes [10]. This process facilitated discussions on VR applications and elicited experts' feedback on potential applications for DMT and critical safety considerations. This was followed by collaborative tagging and discussion around the findings on a Miro board³ (Figure 3). Next, we held an in-depth interview with two experts who possess extensive and diverse professional experience as a therapist or an educator in DMT, and delved deeper into the group interview results, refined application ideas, and validated our findings with them to avoid any potential misunderstandings [4]. Following this, we held an ideation workshop with five experts, where we discussed the storyboard and interaction design for a DMT application in VR. Finally, we conducted another round of in-depth interviews with the same two experts from the initial in-depth interview to finalize the application's guidance content and graphic design.

Based on this iterative process, we identified three key design requirements (DR1, DR2, and DR3) that guided the development of our application. These requirements address the challenges and

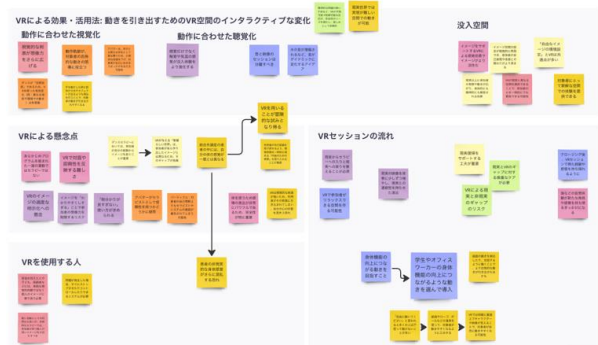


Figure 3: A snapshot of the Miro board used in in-depth interviews.

needs highlighted by the experts, ensuring the application effectively supports DMT practices in a VE.

DR1: Seamless Session Flow. Experts emphasized the importance of smooth transitions into and out of therapy sessions. Traditional DMT practices typically follow three phases—*warm-up*, *exploration*, and *closing*—based on Marian Chace's technique [19], to create a structured flow that eases users into and out of the session. Experts noted that VR environments could make these transitions more abrupt due to the stark contrast between virtual and physical realities, necessitating careful design to bridge this gap. To address this, DR1 focuses on interactions and features that guide users smoothly through session transitions. These include mechanisms for gradual immersion into the virtual space and structured cues to help users reintegrate into the real world after therapy sessions.

DR2: Immersive Imaginative Space. A central component of DMT is imagination, which stimulates creativity and emotional expression. In actual therapy sessions, examples include imagining oneself moving in environments such as underwater, inside honey, or in heaven. Experts emphasized the need for environments that inspire free movement and thematic exploration, allowing unrestricted self-expression. VR was identified as a powerful tool for creating immersive, imaginative spaces that cannot be replicated in physical settings. Examples include environments like being underwater or in a forest, which have the potential to enrich the therapeutic experience by fostering exploration and emotional release.

DR3: Movement-Based Multisensory Feedback. Experts stressed the importance of facilitating spontaneous and improvisational movements, a core aspect of DMT. Therapists use prompts, props, and mirroring techniques in traditional practices to encourage movement. However, in VR environments, dynamic feedback through visual, auditory, and tactile stimuli emerged as an effective alternative to physical props. Experts noted that visual effects like motion trails could help users visualize and reflect on their movements. Dynamic soundscapes responding to the movement were

²<https://jadta.org>

³<https://miro.com/>

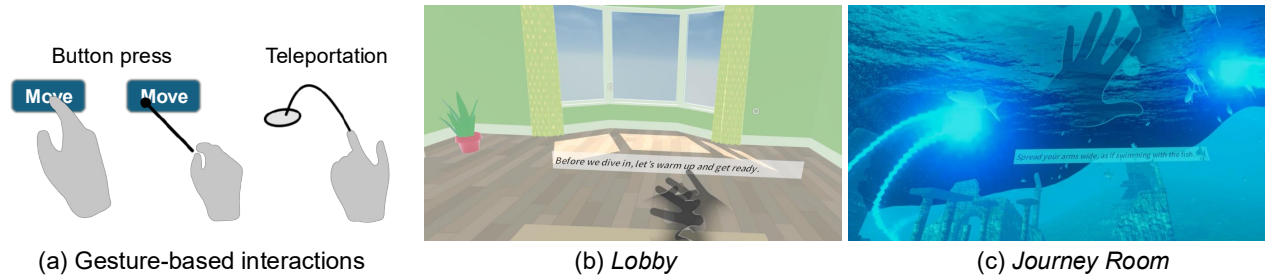


Figure 4: User interaction in *Move & Journey*.

seen as useful for rhythm and engagement, while tactile feedback simulated through virtual interactions could enhance body awareness and emotional connection.

3 SYSTEM

This study developed a prototype of a VR DMT application based on the three design requirements identified in the iterative design process.

3.1 Game Engine and VR Platform

To deploy *Move & Journey*, we used the Oculus Quest 2 VR Head-Mounted Display (HMD) and the Unity 3D game engine. The Oculus Quest platform was selected for its usability, portability and low cost, making it accessible to a wide range of users [2]. In addition, to address seamless and immersive interaction (DR1 and DR2), we used the Meta XR Interaction SDK package⁴ to enable hand-tracking and design for an intuitive interaction with the user interface—see Figure 4 (a) for details on the types of interaction we implemented.

3.2 System Structure and Features

The system is composed of three phases: *warm-up*, *exploration*, and *closing*. The *warm-up* and *closing* phase, each lasting about 2 minutes, take place in a VR space called *Lobby*, which replicates a general space similar to a real room (Figure 4 (b)). This design reduces the gap between real-world and VR sessions. The *warm-up* phase begins with stretching exercises to relax muscles and loosen joints. This process prepares the body for movement while promoting a state of relaxation. The *exploration* phase, lasting about 5 minutes, uses an immersive space called *Journey Room* (Figure 4 (c)). We designed a virtual underwater environment where fish swim around the user based on prior discussions with experts (DR2). At the beginning of this phase, text guidance suggests movements reflecting the rhythm of waves and water flow, encouraging users to walk and move freely within the virtual underwater ocean environment. In addition, to encourage improvisational movement, the system provides real-time visual, auditory, and haptic feedback (DR3). The feedback design is also themed around the ocean, enhancing immersion and stimulating users' imagination (DR2). In the *closing* session, users return to the *Lobby*, where they incorporate slow movements and deep breathing to calm their bodies. This allows them to reflect on the entire session and stabilize their mental and physical states. Additionally, the transitions between *Lobby* and *Journey Room* include guidance such as, “Take a deep breath and keep your eyes closed for five seconds,” to maintain psychological safety and ensure a seamless transition (DR1).

⁴<https://assetstore.unity.com/packages/tools/integration/meta-xr-interaction-sdk-essentials-264559>

3.3 Multisensory Feedback

The *exploration* session incorporates the following three types of feedback designed based on DR3 and current DMT practices:

Visual Feedback. Visual feedback is provided through real-time motion trails that reflect the user's hand movements. These trails, resembling water-like patterns, emerge from both hands and gradually fade within one second. This design draws inspiration from traditional DMT practices, where ribbons or scarves are often used to enhance improvisational movements (Figure 1 (left)).

Auditory Feedback. Auditory feedback plays underwater sound effects in response to the user's hand movements. The system uses an underwater sound⁵ as its audio source. The system calculates hand velocity at 0.1-second intervals using positional data from the controllers. Specifically, it computes the velocity magnitude (v) based on the previous ($\mathbf{P}_{\text{prev}}$) and current ($\mathbf{P}_{\text{current}}$) positions as follows:

$$v = \frac{\|\mathbf{P}_{\text{current}} - \mathbf{P}_{\text{prev}}\|}{t} \quad (1)$$

Using the calculated velocity (v), the system determines a target position (Z) for volume control:

$$Z = a \cdot v + b \quad (2)$$

where a and b are constant values determined through preliminary experiments conducted during the system development. These values were optimized based on the distribution of hand velocities to ensure that faster movements produce higher volumes, while slower movements result in quieter output. To achieve smooth volume transitions, the computed target position Z is linearly interpolated and applied to the AudioManager object, which directly controls the volume level. This approach provides intuitive auditory feedback by dynamically linking hand movement speed to sound intensity. This feature draws inspiration from therapists' verbal cues during traditional DMT sessions (Figure 1 (middle)).

Haptic Feedback. Haptic feedback is delivered through interactions with floating spherical objects. In this phase, the user holds a controller in each hand, enabling them to experience physical sensations through virtual interactions. When the user touches the spheres, they respond with motion and controller vibrations, simulating the physical contact often used in traditional DMT sessions. This design also addresses cultural and gender-related constraints commonly associated with touch-based interactions (Figure 1 (right)).

3.4 Sound and Guidances

The system plays an ocean-themed healing music⁶ in the background. Instead of voice narration, guidances are displayed as text on a GUI—see Figure 4 (b) and (c). The GUI dynamically adjusts its position to remain within the user's field of view at a 3D position

⁵<https://soundeffect-lab.info/>

⁶https://peritune.com/blog/2023/07/21/cerulean_relics/#google_vignette

in the VE, maintaining a consistent distance while moving smoothly. The text transitions automatically, and the GUI disappears once the explanation ends. This design prevents overlap with background music and auditory feedback, ensuring clear visual guidance. Users can follow the text to proceed through the session seamlessly. At the end of each phase, a button appears to allow users to transition to the next room by pressing it (Figure 4 (a)).

3.5 System Evaluation

We conducted an online survey to evaluate the implementation of the design requirements identified in the initial phase and to validate the functionality of *Move & Journey*. Four experts (all female; age 42–73, $M: 59.4$, $SD: 15.9$) with 12–24 ($M: 20.0$, $SD: 5.65$) years of experience in DMT instruction, who had participated during the development phase, were involved. They applied their expertise to identify any undesirable or missing features and interactions [22].

We asked the experts to watch a 2D recording captured from the HMD perspective of an individual using our application. After viewing the video, they completed an online questionnaire evaluating the system based on *seamless session flow* (DR1), *immersive imaginative space* (DR2), and *movement-based multisensory feedback*, including visual, auditory, and haptic elements (DR3) using a 5-point Likert scale (1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree). We also included open-ended questions to allow experts to provide additional feedback.

Figure 5 shows the mean scores for each questionnaire item, with error bars indicating standard errors. Overall, experts rated the system as an effective stress-relief tool ($M: 4.00$, $SD: 0$), emphasizing its ability to evoke enjoyment and positive emotions. Regarding DR1, experts highly rated the smooth flow of the session ($M: 3.75$, $SD: 0.50$) and particularly appreciated the guidance that encouraged sensory awareness ($M: 4.00$, $SD: 0$). However, an expert suggested “*reducing explanatory instructions and guidance to allow users to actively select information from the virtual environment.*” For DR2, the immersive VE ($M: 3.75$, $SD: 0.50$) and imaginative design ($M: 4.25$, $SD: 0.50$) received positive evaluations, although an expert commented that “*the white objects in the underwater scene evoked an impression of ice and felt cold,*” recommending warmer visual designs. Additionally, another expert proposed “*incorporating slower tempos, closer to resting heart rates, and variations in musical elements such as tempo, tonality, and dynamics to enhance depth and satisfaction.*” With respect to DR3, experts responded positively to visual ($M: 4.25$, $SD: 0.50$), auditory ($M: 4.00$, $SD: 0.00$), and tactile feedback ($M: 3.75$, $SD: 0.50$), with visual feedback receiving particularly high ratings. The vibration-based tactile feedback was also described as “*interesting and effective in generating positive emotions.*” However, some experts suggested “*integrating tactile feedback with user actions, such as enabling upward movement toward light by mimicking swimming motions,*” to strengthen immersion and promote positive imagery.

4 DISCUSSION AND FUTURE WORK

Building upon the promising results of this study, we plan to further refine the system to enhance its effectiveness as a stress-relief application. While the current design demonstrates potential through its sensory-guided structure, immersive features, and multimodal feedback, areas for improvement have been identified, including refining narration, adjusting music design, and enhancing interaction mechanisms. Incorporating these insights is expected to increase user satisfaction and deepen relaxation effects.

We are working on completing the detailed system design and implementation. Following this, we plan to conduct large-scale user experiments using the finalized system, targeting healthy students and office workers. In the user experiments, we will evaluate the system’s effectiveness in stress relief using methods similar to other VR therapy studies [6]. Specifically, we will employ the Paced Auditory

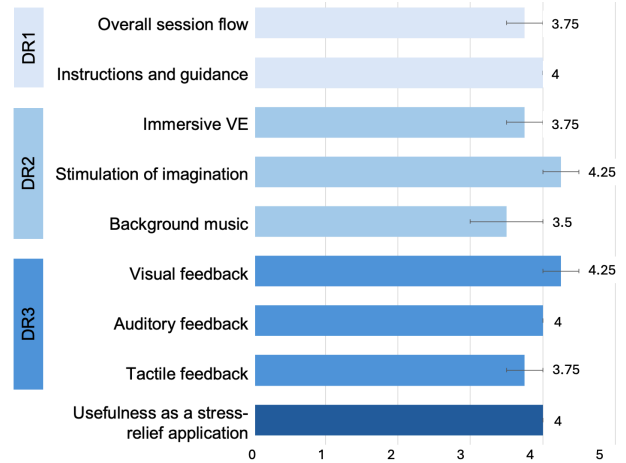


Figure 5: Evaluation questionnaire results.

Serial Addition Test (PASAT) as a stress inducement test [21] and apply stress assessment methods to verify the effects of the system. We will also collect detailed user feedback through a post-experiment questionnaire to refine the system further. While this study primarily focused on the experts’ perspectives, there are numerous aspects to be explored in VE. For instance, further investigations are needed regarding feedback design, avatar presence, and types of user perspectives. By addressing these aspects, we aim to create a more effective stress-relief application that leverages VR technologies to promote relaxation and well-being across diverse user groups.

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